Applied Information Extraction in Python

Today's goal • 3/3 items

Module 1
Information Extraction

Module 2
Named Entity Recognition (NER)

Module 3
Neural Network Models

What Is Deep Learning?

Introduction to Deep Learning
Video • 4 min

Understanding Deep Learning: A Shift From Rules to Representation
Reading • 10 min

Knowledge Check: What Is Deep Learning?
Practice Assignment • 10 min

What Are Neural Network Models?

Neural Network Models
Video • 7 min

Activation Functions: How Deep Learning Models Make Decisions
Reading • 10 min

Knowledge Check: What Are Neural Network Models?
Practice Assignment • 10 min

Deeper Neural Networks

Deep Neural Network Models
Video • 6 min

Understanding Deep Neural Network Models: How Depth Enables Learning at Scale
Reading • 10 min

Knowledge Check: Deeper Neural Networks
Practice Assignment • 10 min

How to Build IE Models Using DNNs

Building an Information Extraction Pipeline with BiLSTMs and CRFs
Reading • 10 min

Jupyter Notebook Practice on Training LSTMs
Lab • 1h

Demo: Configuring the Bi-Directional LSTM
Video • 4 min

Module Wrap-Up & Assignment

Assignment 3 Introduction: Building an Information Extraction Pipeline with BiLSTMs and CRFs
Video • 3 min

Build an Information Extraction Pipeline using Deep Neural Networks
Programming Assignment • 3h

Module 3 Assignment
Graded Assignment • 30 min

Module 4
Transformers Transform Information Extraction

Understanding Deep Learning: A Shift From Rules to Representation

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From Traditional Machine Learning (ML) to Deep Learning

We've reached a turning point in this course: transitioning from traditional machine learning to deep learning. In earlier modules, you saw how models like HMMs or CRFs relied on engineered features, probabilistic transitions, and explicit labeling rules.

But deep learning changes the game.

Instead of learning from handcrafted features, deep learning models learn what the features should be, automatically, from raw data. That shift has made deep learning the state-of-the-art approach across domains like speech recognition, image processing, and clinical text mining.

Let's explore why that matters, and how deep learning differs from traditional methods.

Traditional Machine Learning

In traditional models (like logistic regression, SVMs, or CRFs), the workflow typically looks like this:

Input Text → Feature Engineering → Classifier → Output

The success of the model depends heavily on:

- How well you design features (e.g., word position, token shape, part-of-speech)
- How effectively the model uses those features

This pipeline works well for structured data, but breaks down with unstructured input like free text, medical notes, or images. Feature engineering becomes:

- Tedious
- Task-specific
- Brittle (i.e., easily broken by small domain shifts)

Deep Learning:

Deep learning removes the manual feature design step. Instead, it learns to extract representations directly from the raw data during training.

The new pipeline becomes:

Raw Input → Neural Network (Feature + Classifier in one) → Output

In this setup, the neural network:

- Automatically identifies useful features
- Learns to assign importance via weights
- Optimizes both representation and classification together

This is called **representation learning**, one of the key reasons why deep learning models often outperform older methods, especially on complex or messy data.

Benefits of Deep Learning

- No manual feature engineering
- Highly flexible for text, speech, images, time series, etc.
- Scales well with large datasets
- Enables state-of-the-art performance in tasks like:
  - Named Entity Recognition (NER)
  - Adverse event detection
  - Clinical concept extraction

Examples:

- BERT and GPT are deep learning models built on transformer architecture.
- In healthcare, deep learning powers tools that extract diagnoses, medications, or conditions from clinical notes, without needing a pre-defined vocabulary list.

Limitations of Deep Learning

Despite its strengths, deep learning has trade-offs:

- Low interpretability It's harder to explain why a model made a specific prediction.
- Needs lots of data Deep networks overfit on small datasets.
- High computational cost Training requires significant hardware resources, especially for large architectures.

Still, these are active research areas, and newer models are improving both interpretability and efficiency.

Deep Learning vs. Traditional Machine Learning: A Quick Comparison

Aspect	Traditional ML	Deep Learning
Feature Design	Manual (you do it)	Learned by the model
Data Requirement	Works with small data	Needs large datasets
Interpretability	High	Low
Performance	Good	Often better
Pipeline	Feature extraction → Classifier	All-in-one (end-to-end)

Suggested Readings & Tools

- Goodfellow, Bengio, & Courville – Deep Learning
  - Chapter 1 (Introduction) and Chapter 6 (Deep Feedforward Networks) are especially relevant.
- 3Blue1Brown: Neural Network Intuition (YouTube)

Reflection Prompt

In traditional ML, the model relies on the features you give it. In deep learning, the model creates its own.

How might that change your role as a data scientist? In what situations would you still prefer traditional methods over deep learning?

Assignment Preview

As you begin Module 3's hands-on tasks, you'll explore neural networks from the ground up, starting with a single neuron and scaling up to deep architectures.

You're now building models that see the world more like humans do, not by rules, but by experience.

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